Regime-Aware Order Book Forecasting Under Distribution Shift

Perceptron Hackathon 2026 — Theme 3: Markets & Trading Intelligence — Problem 4

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1. Problem and Theme Chosen

Theme 3, Problem 4. In poker, I compress observed opponent actions into a behavioural model rather than retaining every hand. Likewise, order-book forecasting compresses noisy events into states that preserve information useful for predicting price moves.

2. Understanding of the Challenge

The task is to forecast short-term mid-price moves and detect when training conditions no longer hold. *Covariate shift* changes feature distributions; *concept drift* changes the feature–future-return relationship. Useful regimes must persist (AAABBB, not ABABAB) and predict; clustering quality alone is insufficient.

3. Proposed Approach

Data. LOBSTER develops the pipeline; self-recorded multi-day crypto L2 data test temporal generalization. Bar types (time, event-count, volume or dollar) are selected by inspection (López de Prado, 2018).

Stage 1 — Representation selection. Each window $W_t \in \mathbb{R}^{w \times N}$ contains standardized book imbalance, OFI, spread, depth, trade imbalance and past returns. Scalers are fitted on training and frozen on validation/test. Manual candidates encode each feature with two or three fixed, predefined statistics (mean and standard deviation, plus one of latest value, maximum, skewness or lagged slope), keeping the manual baseline competitive; they are compared with learned attention pooling under the protocol below. Selection has two steps: (1) normalized reconstruction loss, using a shared target, common latent dimension and comparable-capacity decoder (the manual vector is projected), screens information preservation rather than predictive utility; (2) surviving candidates are compared on cluster occupancy, pre-filter persistence (mean duration/switch rate), silhouette separation and regime-conditioned return/volatility distributions. Walk-forward forecasts alone determine predictive value. We test $K \in \{2, 3, 4\}$ with K-means; the causal filter is used in predictor training and inference, but excluded from structural persistence metrics.

Stage 2 — Unknown states and shifts. For latent vector z_t, pointwise novelty is $s_t = \min_k \|z_t - c_k\|_2/\rho_k$; centroids and dispersions are fitted on training data, with τ calibrated on chronological validation. Scores above τ or regimes with insufficient training support are unknown. A rolling two-sample KS test compares recent and reference scores; block-bootstrap validation thresholds set alarms, and false alarms/hour are reported. Predictions are withheld while either condition is active and resume once both clear. Persistent shifts therefore cause sustained abstention; recovery through new regimes remains offline, not online. Delayed log-loss degradation flags possible concept drift but triggers no automatic action.

Stage 3 — Prediction. Development tests h = 5–60 seconds; one primary horizon is locked before the final test. The target $y_{t+h} \in \{-1, 0, +1\}$ classifies $r_{t,h} = \log(p_{t+h}/p_t)$ relative to $\pm\kappa\hat{\sigma}_{t,h}$, a causal volatility estimate for that horizon. Inputs $x_t = (I_t, Q_t, C_t)$ are: top-L_b book-depth imbalance $I_t = \frac{1}{n}\sum_{i=t-n+1}^{t} \frac{V_i^b - V_i^a}{V_i^b + V_i^a}$; aggressive-trade imbalance $Q_t = (B_t - S_t)/(B_t + S_t)$; and cumulative log-return $C_t = \log(p_t/p_{t-n})$. Logistic regression, Random Forest and XGBoost are compared as global and regime-conditioned models with limited validation tuning. Features are causal; labels whose horizons cross train/validation/test boundaries or regime segments are purged. Class probabilities are primary; training-period class means provide secondary expected returns.

4. Minimum Viable System

The MVP includes both aggregation families, K-means, distance-based abstention, rolling KS monitoring and three classifiers. Logistic regression is the reference; Random Forest and XGBoost share features, splits and limited tuning. Unknown windows are logged, not retrained online.

5. Stretch Ambitions

Recorded unknown observations will be analyzed offline for similarity, recurrence and distinct return behaviour. If a coherent candidate regime recurs and proves useful on subsequent data, a later version could add it and retrain a dedicated head offline. Predefined cross-feature statistics (e.g., OFI–return correlation and book-versus-trade imbalance differences) test whether interactions improve regimes; a VAE tests latent regularization; neural heads and latency benchmarking test predictive flexibility and deployment cost.

6. Novelty and Differentiation

Our contribution is a flexible, multi-step evaluation framework for aggregation functions, rather than a new pooling operator. It accommodates fixed manual encodings and learned attention: reconstruction loss screens information preservation, regime diagnostics assess structure, and walk-forward tests establish predictive value. We also distinguish pointwise novelty from collective score-distribution shifts and evaluate abstention against random and volatility-based baselines at matched coverage.

7. Evaluation Plan

Purged walk-forward splits use a locked test period; encoders, scalers and centroids are fitted on training data, thresholds on validation. Log-loss is primary, with directional hit ratio on accepted non-neutral outcomes, Spearman IC and block-bootstrap intervals. Controls include global linear and nonlinear models, testing whether regime conditioning adds value beyond model complexity, plus regime-specific class priors, block-permuted labels and time-of-day regimes. Shift metrics are detection delay, false alarms per hour and performance around shifts. Abstention is assessed by coverage and matched-coverage log-loss versus random and volatility-based baselines.

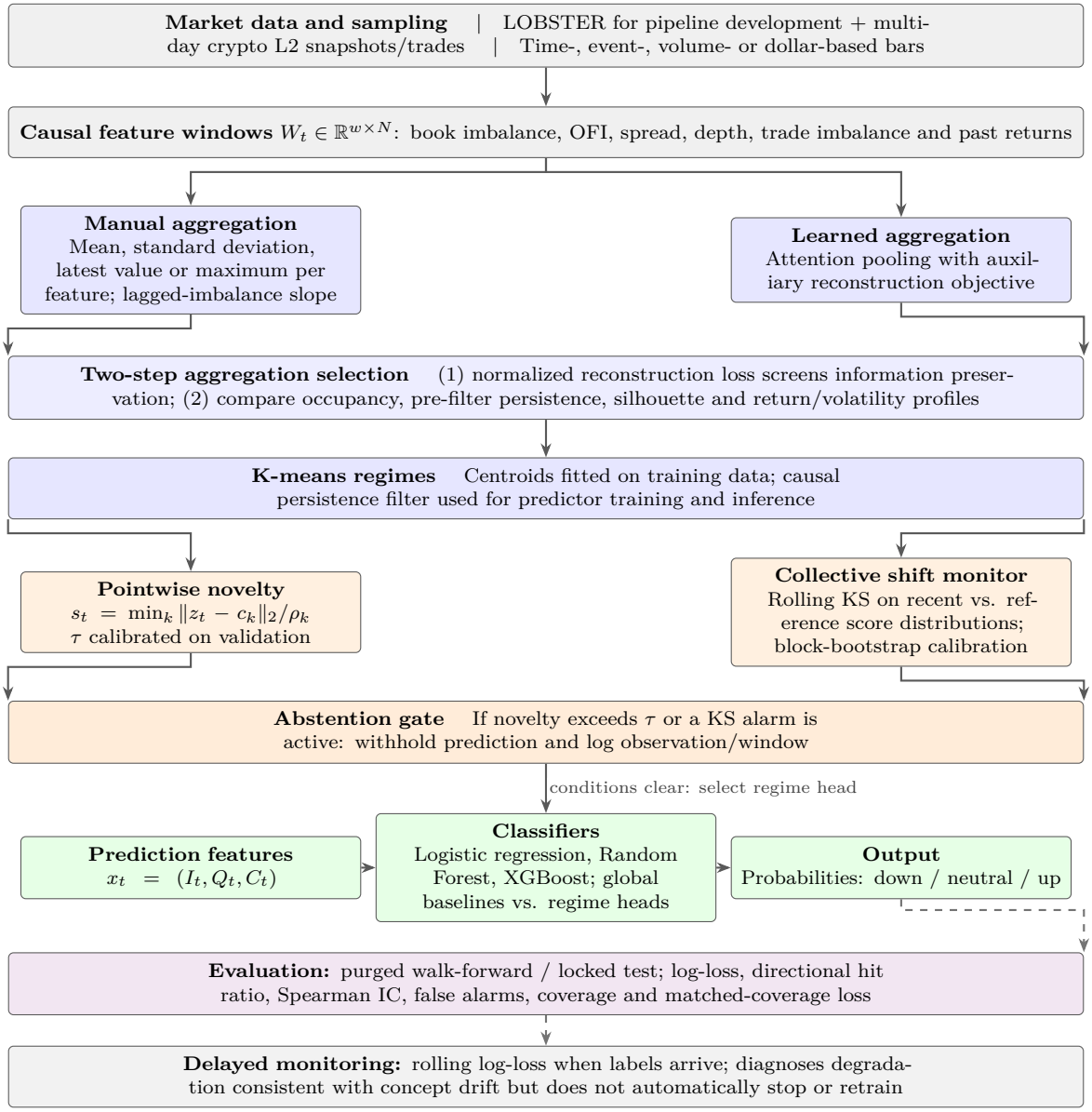
8. Feasibility and Risks

Work starts 15 October 2026; six weeks cover data/baselines, representation/detection/prediction and evaluation/polish, with a buffer before the 1 December Midway Check. Risks include few natural shifts, intraday-only clusters and weak gains; multi-day crypto, time-of-day controls and separately reported controlled shifts address them. Second-year engineering student working on independent ML and quantitative-finance projects, including tick-level order-flow analysis, regime research and Rust-based derivatives pricing. Selected work: GitHub and Medium.

Reference: López de Prado, M. (2018). *Advances in Financial Machine Learning.* Wiley.

System Overview

Regime-aware forecasting under distribution shift



Solid arrows: inference and detection flow. Dashed arrows: evaluation and delayed predictive monitoring.